

Peter Majors

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Backend engineer specializing in Python, distributed systems, and LLM-native applications. Builds scalable financial reporting, data, and automation platforms that accelerate business decision-making.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, R, C++

Frameworks & Libraries: FastAPI, Flask, Celery, Pandas, Scikit-learn, Spark, LangChain, llama.cpp

Cloud & Databases: Amazon Web Services, PostgreSQL, MongoDB, Redis, Databricks

DevOps & Tools: Docker, Kubernetes, GitHub Actions, Datadog, Snyk, Azure DevOps

RELEVANT EXPERIENCE

Marsh & McLennan Companies, Inc.

New York, New York

Senior Application Development Analyst - Autonomous Finance

Jul. 2023 – Present

- Engineered financial reporting platform for **\$20B** P&L generating **500+** GAAP & regional decks for month-end close
- Designed FastAPI/PostgreSQL backend enabling Okta SSO, in-app messaging, approval workflows, and SharePoint integrations
- Architected async Celery-based ETL generating **4,000+** reports hourly from Oracle Essbase with automated variance commentary
- Implemented distributed Redis pub/sub and WebSocket architecture supporting real-time collaboration for **330+** global controllers
- Led **8** production releases, coordinating testing with stakeholders and ensuring system reliability through Datadog and Flower
- Built LangGraph agent to query EPM data and traverse business taxonomies via NetworkX graphs, supporting chat interface for FP&A research; developed evals pipeline to quantify impact of context engineering, improving accuracy **30%** over baseline

Finance Innovation & Technology Intern

Jun. 2022 – Aug. 2022

- Built Flask REST API with JavaScript client performing ASC 606-compliant broker service amortization calculations; partnered with controllership teams on product design

Proskauer Rose, LLP

New York, New York

Accounting Intern

Jun. 2021 – Jul. 2021

- Automated journal entry and reconciliation workflows using Excel Macros, expediting two month-end close cycles

LEADERSHIP & COMMUNITY

Fordham Sports Analytics Society - Fordham's Best New Student Organization (2021)

Bronx, New York

Founder, President

Oct. 2020 – May 2023

- Founded Fordham's first sports analytics club and grew active membership to **30+** students through weekly research presentations, data science workshops, and **25+** interviews with MLB, NFL, and NBA analysts and executives
- Launched and edited research blog featuring **36** data-driven sports articles averaging **300+** views, increasing club visibility
- Partnered with Fordham Athletics to establish University's first student-run baseball scouting department and leverage analytics platforms to provide opponent analysis for football, women's basketball, and women's soccer

Cystic Fibrosis Foundation - Tomorrow's Leaders

New York, New York

Volunteer & Professional Development Council Member

Oct. 2023 – Present

- Raised **\$5K** for CF research by completing the 2024 NYC Half-Marathon with the Breathe Team
- Expanded mission awareness and donor base by recruiting colleagues and peers to bi-monthly networking events

CASE COMPETITIONS

March Data Crunch Madness - Fordham GSAS

Bronx, New York

Team Lead - Placed 3rd of 22

Jan. 2022 – Mar. 2022

- Led development of an XGBoost classification model to predict 2022 NCAA Men's March Madness matchup outcomes
- Introduced an "Adjusted Seed Gap" feature based on KenPom's AdjEM to quantify team over/under-seeding, achieving prediction accuracy of **65%** for matchups with ≤ 5 seed differential

Syracuse Football Analytics Blitz

Bronx, New York

Team Member - Placed Top 4 of 17

Feb. 2022

- Analyzed PFF and nflfastR play-by-play data with Pandas to identify offensive tendencies for the 2021 Kansas City Chiefs
- Developed Scikit-learn logistic regression models to forecast play calls (pass vs. run, play-action, and motion) for 5 game states to guide coverage and personnel recommendations aimed at minimizing EPA/play

EDUCATION

Fordham University, Gabelli School of Business

Bronx, New York

Bachelor of Science in Accounting Information Systems, Minor in Computer Science | GPA: 3.8/4.0

Sep. 2019 – May 2023

Academic Awards: Summa Cum Laude, Gabelli School of Business Dean's List

Coursework: Advanced Python for Business, Internet & Web Programming, Database Systems, Cyber Secure Networks